Molecular Digital Product Passport: An AI-Supported Decision System for Textile Characterisation

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Digital Product Passports (DPPs) can make textile composition information available for reuse, recycling, and compliance, but access to that information depends on a physical carrier such as a QR code or RFID tag. When that carrier is damaged or removed, the fabric must instead be characterised directly from the material itself. This thesis develops an AI-supported decision-support prototype for tagless fabric characterisation in a DPP context. The prototype supports both reference comparison and composition estimation by combining microscopy image embeddings with NIR spectra through late fusion, while keeping per-test outputs, combined outputs, and uncertainty or conflict warnings visible to textile specialists.

The evaluation covers the image classifier, the image and spectra representations, and the final composition-estimation output. It compares two CNN iterations, checks representation similarity on validation and test data, uses Grad-CAM heatmaps and a validation-image inference-scaling test, and evaluates composition estimation on the validation and locked test splits. Fine-tuning the final DenseNet201 feature block improves the validation weighted-F1 to 0.906, with a weighted-F1 of 0.750 on the locked test split, above the majority-class baseline. For composition estimation, spectra-only retrieval performs best, image-only retrieval improves over the training-mean baseline, and the combined image and spectra estimates fall between the two individual modalities. Next steps include testing on larger datasets, calibrating confidence estimates, normalising reference-comparison scores, and evaluating the prototype with textile specialists.

Additional Key Words and Phrases: Computer vision, Digital Product Passport, textile characterisation, multimodal fusion

1 Introduction

Global warming is causing rapid and widespread changes in the atmosphere, ocean, biosphere, and cryosphere [24]. Among the many contributors to this crisis, the textiles industry stands out as a major source of global carbon emissions [15, 36]. The apparel and footwear sectors alone are estimated to generate 8 to 10% of annual global carbon emissions, and these emissions have been projected to account for roughly 25% of the carbon budget by 2050. Synthetic fibres are especially problematic: producing a polyester t-shirt is estimated to contribute 5.5 kg CO2-equivalent (CO2-e), which is 3.4 kg CO2-e more than producing a comparable cotton one.

Several policy initiatives have been launched to reduce the climate impact of textiles and other industries. One example is the European Green Deal, approved in 2020 [5] and made legally binding through the European Climate Law [19]. This law requires all member states to contribute to the goal of making Europe the world's first climate-neutral continent by 2050. A key target is to reduce greenhouse gas emissions (GHG) by at least 55% compared with 1990 levels by 2030.

The European Green Deal includes initiatives across the economy, energy, industry, and transport [5]. One of these is the Circular Economy Action Plan, which contains legislative and non-legislative measures addressing the full life cycle of products, including batteries and textiles [1, 2]. Part of this action plan is the legally binding Ecodesign for Sustainable Products Regulation (ESPR) [12]. The ESPR supersedes Directive 2009/125/EC, which focused on energy-related products such as washing machines and insulation [3]. The new regulation is broader and targets almost all non-consumable products.

To shape the practical implementation of these regulations, the ESPR uses a co-design methodology involving experts and stakeholders. For textiles, this process informs requirements for more durable, sustainable, and repairable products. In this way, the ESPR supports a shift from an unsustainable linear model of take, make, use, and dispose toward a circular economy, where products are designed to last longer and be reused or recycled [1].

An important mechanism introduced by the ESPR is the Digital Product Passport (DPP). A DPP is a digital record intended to make product information more accessible for sustainability, circularity, and compliance purposes. Rather than defining one fixed passport for all products, the ESPR allows DPP requirements to be specified for particular product groups through delegated acts. For textiles, these requirements can include information about

durability, material composition and origin, repair activities, recycling capabilities, and life-cycle environmental impacts [12].

The future implementation of DPPs is expected to face several challenges, including the physical medium used to access the information [51]. A QR code or RFID chip could reference a DPP's digital record, but these physical carriers can be damaged or removed by consumers. Without reliable physical tags, alternative tagless methods are needed to characterise textiles for tasks such as recycling.

To address this gap, the Molecular Digital Product Passport (M-DPP) project, supported by the Dutch Research Council (NWO), was initiated. As part of this project, the following research question was posed: "How can an Artificial Intelligence (AI) supported decision system using multimodal tests contribute to the characterisation of fibres while handling uncertainty and supporting textile specialists in a Digital Product Passport context?". This research question is addressed through the following sub-questions:

(1) What fabric characterisation tests are available and how accurate are they?
(2) How can test results be combined into a shared representation while remaining robust to missing tests and weighting tests according to their reliability?
(3) Which architecture is the most appropriate for the proposed multimodal decision system, and how should it be evaluated?

2 Background

Textiles are structured hierarchically: fibres are spun into yarns, yarns are woven or knitted into fabrics, and fabrics are cut and sewn into garments [56]. This project focuses on fabric samples rather than whole garments, because garments can contain multiple materials. At fabric level, textile composition can be assessed through physical and chemical analysis [32]. This section discusses characterisation methods and why no single method fully characterises a fabric. It then covers deep learning, test-representation fusion, AI suitability, ethical and societal considerations, and stakeholders.

2.1 Fabric Characterisation Methods

Fabric characterisation methods target different properties, including chemical composition [32], molecular structure [41], mechanical behaviour [22], and visual appearance [30]. Some methods can identify certain fibre types with high confidence, but reliability depends on blend composition and sample conditions [48]. When composition is unknown, the most suitable test may be unclear or unavailable. Multiple methods can therefore provide alternative evidence and indicate uncertainty when results disagree.

Characterisation methods can be destructive [32] or non-destructive [41]. Destructive examples include burning and solubility tests. Burning tests can distinguish synthetic fibres, which melt into hard beads, from natural fibres, which generally combust into soft ash or char. Solubility tests expose fibres to chemical solvents, where each fibre type dissolves or remains intact depending on its chemical composition, supporting fibre identification.

Non-destructive methods avoid destroying the sample. A common example is FT-IR Spectroscopy, which measures how textiles absorb infrared radiation [41]. Because molecular structure determines which wavelengths are absorbed most strongly, the resulting spectrum acts as a fingerprint that can be compared with reference samples to identify fibre types [20]. However, FT-IR can struggle to distinguish materials with similar molecular structures. Polyethylene terephthalate and polybutylene terephthalate, for example, are both polyesters and produce very similar spectra, so additional methods can improve reliability in such cases [41].

Raman Spectroscopy is another spectrographic method for fabric characterisation [42]. It directs a laser beam at the fabric and measures the scattered light to produce a molecular spectrum. Because FT-IR measures infrared absorption and Raman measures scattered laser light, each technique can detect molecular features the other misses [42].

Mechanical tests also differ in how much they affect the sample. Bend tests measure the length at which a fabric bends to a specific angle, indicating stiffness without necessarily destroying the sample [22]. Tensile tests are destructive because they stretch a fabric until it breaks while recording maximum force and elongation before failure [13].

2.2 Deep learning for textile characterisation

Interpreting spectrographic and physical test results requires specialised domain knowledge. Deep learning approaches can serve as an alternative or supplement to these characterisation methods. Convolutional Neural Networks (CNNs), for example, have a low barrier at use time because they only require an image as input. They still need labelled data and machine-learning expertise for training, configuration, and evaluation. CNNs can automatically extract textile-image features and have been applied to defect detection [30], knitted fabric structure recognition [57], and post-consumer textile waste classification [45].

Other approaches, such as autoencoders [23], can learn compact fabric-image representations by reconstructing their input through a low-dimensional bottleneck. However, because the training objective is reconstruction, the embeddings are not necessarily optimised to separate fibre types. They preserve the information most useful for rebuilding the image.

Supervised Contrastive Learning (SCL) [33] trains a model to produce similar embeddings for samples of the same class and dissimilar embeddings for samples of different classes. This makes it more suitable for fibre-type similarity comparison than autoencoders because it optimises the feature space around class distinctions. However, SCL relies on large batch sizes for diverse in-batch negatives; small batch sizes introduce gradient bias, which can weaken the learned representations [25]. Dataset size and batch composition are therefore important when deciding whether contrastive learning is suitable for textile image representations.

Because CNNs have also been used for woven fabric pattern classification [29] and textile-fibre classification [45], the next subsection focuses on CNN architectures reported for textile classification tasks.

2.3 CNN Architectures for Textile Classification

CNN architectures differ in how they extract visual features from images. In classification models, the layers before the final classifier can also be used as feature extractors, producing embeddings that represent visual properties of the input image. Table 1 summarises three image-based textile classification studies across different datasets and model designs.

Table 1. Comparison of CNN architectures used for textile classification tasks.

Architecture	**Dataset / task**	**Reported performance and relevance**
DenseNet201 + SVM (RBF)	Self-made mobile-image dataset with 14 commercial fabric classes and 48 images per class (672 total); 80/20 train-test split with 10-fold cross-validation	Barros et al. compared several CNN feature extractors and reported the best result with DenseNet201 features and an SVM (RBF) classifier: 94.35% accuracy and a 94.30% F1-score. The study is relevant because it identifies fabrics from images, although the images are mobile-camera photographs [27].
ResNet50	TextileNet-fabric subset with RGB garment images and 27 fabric labels	TextileNet reports a ResNet50 baseline accuracy of 65.28±0.67%; F1-score was not reported. This provides an image-based CNN comparison for textile material identification, although the data consists of RGB garment images rather than fabric-level images [58].
FabricNet (CNN ensemble)	Dataset with 7,553 fabric and garment images across 50 fibre labels, including 9 identified non-blended fabric types	FabricNet is a CNN ensemble for image-based fibre recognition, reporting 84% accuracy and a 90% F1-score. Ohi et al. first compared established CNN architectures, then built FabricNet as an ensemble model for multi-label fibre prediction. It is relevant because it predicts fibre labels from fabric images, although the dataset also contains garment images [39].

2.4 Test Representations

Textile characterisation methods produce different outputs. Some produce feature embeddings or classifications [30], others produce spectra [42], and others produce measurements such as bending length or elongation at break [13]. These outputs therefore need a representation and comparison strategy.

Because these outputs differ in format and meaning, combining them requires a way to compare evidence across tests. Combining multiple test outputs is generally called multimodal fusion [35]. In the M-DPP project, not all possible tests are known in advance, and not every test will be relevant or available for each fabric [35]. The system therefore needs an approach that can work with whichever tests are available. Stahlschmidt et al. describe three main modality, or test, fusion strategies [54]:

(1) **Early fusion:** All test outputs are concatenated into a single vector before processing. For example, a spectrum $[0.1, 0.4, 0.9]$ and measurements $[12.3, 0.8]$ are combined into $[0.1, 0.4, 0.9, 12.3, 0.8]$, which is then used as model input.
(2) **Intermediate fusion:** Each test is first processed by its own sub-network to produce a compact feature vector. These vectors are then combined. For example, a spectrum sub-network might produce $[0.3, 0.7]$ and a measurement sub-network $[0.5, 0.2]$, which are merged into $[0.3, 0.7, 0.5, 0.2]$ for the final layers.
(3) **Late fusion:** Each test is processed independently and produces its own prediction. These predictions are then combined into a final output. For example, one model might output 80% wool and another 60% wool, after which the scores are merged into a single prediction.

Late fusion requires an aggregation strategy to combine the independent predictions from each test-specific model. Stahlschmidt et al. distinguish three main methods for this purpose [54]:

(1) **Averaging:** Each test prediction is given equal weight and the predictions are averaged. For example, scores of 0.8 and 0.6 become 0.7.
(2) **Weighting:** Each test is assigned a weight reflecting its reliability or relevance. For example, one test output might receive a weight of 0.7 and another 0.3, producing a weighted final prediction.
(3) **Meta-learning:** Predictions from each test are used as input to, for example, a machine-learning model, which learns how to combine them. It might, for instance, learn to rely more heavily on one test when another produces uncertain predictions.

Each aggregation method has trade-offs [54]. Averaging is simple and needs no additional weighting or training, but works best when tests are equally reliable. Weighting is slightly more complex, but can include domain knowledge about test relevance or reliability and can handle missing tests. Meta-learning is the most complex because it requires a separate model. With enough data it may improve performance, but changes in available modalities may require retraining.

When test weights depend on expert judgement, Multiple Criteria Decision Making (MCDM) methods can be used. The Analytic Hierarchy Process (AHP) assigns weights through pairwise comparisons, but requires $n(n-1)/2$ comparisons, which becomes impractical as the number of tests grows. The Best-Worst Method (BWM) reduces this burden: the expert identifies the most and least important tests and compares the remaining tests against these two reference points, requiring $2n-3$ comparisons [44, 47]. For 10 tests, this reduces the task from 45 to 17 comparisons.

2.5 AI suitability justification

The suitability of AI follows from the difficulty of fabric characterisation. Different tests capture different properties and vary in how easy they are to perform and interpret, so characterisation requires domain knowledge [22, 32, 41]. A single test can sometimes be sufficient, but spectrographic, mechanical, and visual tests each provide only part of the picture, so their outcomes may need to be interpreted together [35]. This is especially difficult when results

are ambiguous or contradictory. A system that structures and compares these outcomes can therefore support interpretation.

Many characterisation methods also require specialised equipment or expertise. Image data offers a practical additional source of information, but its patterns are difficult to interpret manually. AI can learn from labelled examples and extract useful information from image data [26, 30, 39]. It therefore contributes additional evidence, while existing tests remain necessary for a broader and more reliable assessment.

In this project, the AI component focuses primarily on the image modality. A CNN can identify visual patterns in microscopy images that are difficult to capture through fixed rules alone. Within the broader system, this evidence supports two forms of interpretation: comparison with a specific reference fabric when available, and similarity-based interpretation using stored fabrics with known compositions when no direct reference is available. The first can support checks of whether a claimed composition matches the tested fabric; the second can support cases where composition must be inferred through comparison with known samples.

AI should therefore be understood as one part of a broader decision-support process. It provides one source of evidence within an interpretation that depends on multiple test outcomes. When outcomes are uncertain, incomplete, or conflicting, this should be made clear to the user rather than hidden behind a single definitive conclusion.

2.6 Ethical and Societal Considerations

This project supports professionals involved in textile analysis and validation by helping them interpret test results. Its outputs are intended to inform human judgement rather than replace it. This human-in-the-loop design aligns with the EU AI Act's emphasis on human oversight and mitigation of automation risks [4]. However, automation bias remains a risk: users may place too much trust in outputs because they appear consistent or data-driven. Since the system can make mistakes, users should be made aware that its outputs are not definitive and require human judgement.

The impact differs between actors because the system supports different decisions in the textile chain. Textile specialists can inspect whether test results point in the same direction through per-test outputs, similarity scores, and conflict warnings. Recyclers or sorters can use composition estimates to support routing decisions for unlabelled fabrics, while uncertainty cues show when evidence is weak or incomplete. Sellers or organisations preparing product information can check whether claimed fibre compositions are supported before the information is used in a Digital Product Passport. Consumers are affected more indirectly. They are unlikely to inspect test results themselves, but validation or certification processes can use this evidence to check fibre-composition claims behind product information. In this way, the system can support trust without requiring consumers to understand the technical characterisation process.

In a DPP context, these decisions also have broader societal effects. Better fabric characterisation can help check whether claimed product compositions match the tested material, strengthening trust and reducing room for misleading sustainability or composition claims. It can also improve sorting, reuse, and recycling decisions, where fibre composition affects which materials can be processed together and which recovery routes remain feasible. In that sense, improved characterisation can contribute to circular-economy goals by helping textiles remain in use for longer and by improving end-of-life routing. At the same time, incorrect composition estimates may negatively affect validation, sorting, or material handling decisions. Making uncertainty and conflicting evidence clear is therefore ethically important.

2.7 Stakeholder Analysis

The project proposal identifies four main stakeholder groups [43]:

- **HvA Fashion Research & Technology:** Project lead; responsible for participatory co-design and stakeholder engagement.
- **HvA Responsible IT:** Technical infrastructure lead; responsible for developing the architecture and open-source APIs.
- **HAN Biobased Innovations:** Molecular analysis lead; responsible for developing and validating molecular identification methods for textile fibre characterisation.
- **Consortium partners:** Industry, technology, and dissemination partners.

These stakeholders have different levels of power and interest within the thesis scope, as shown in Figure 1. Responsible IT has the most power because it leads the technical infrastructure. Fashion Research & Technology has medium power because it leads the broader project and can make requests to Responsible IT. HAN Biobased Innovations and the consortium partners have high interest but lower direct influence on the thesis requirements. The matrix is based on a meeting with Responsible IT, where stakeholder roles and their relative power and interest were discussed.

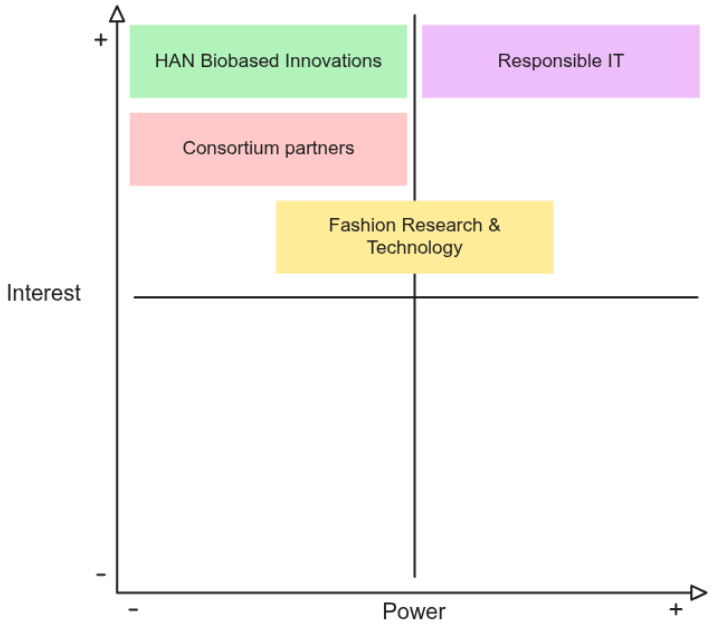


Fig. 1. Power–interest matrix of the project stakeholders, showing their relative power and interest.

3 Methodology

This section describes the prototype requirements, concept design, data preparation, model architecture, and evaluation method.

3.1 Requirements

Table 2 lists the functional (F), ethical (E), legal (L), organisational (O), and technical (T) requirements. Because fabric characterisation can use several test types, the prototype supports multiple test outputs, represented by microscopy images and NIR spectra, so the requirements cover both their fusion and their evaluation.

Table 2. Requirements.

ID	Requirement	Justification	Definition of Done
F1	The system supports reference comparison and composition estimation.	Responsible IT defined these modes: one compares against a selected reference fabric, while the other estimates composition from stored fabrics with known fibre-composition metadata.	The system returns either a reference similarity score or a composition estimate.
F2	The system combines multiple test outputs into one overall output.	A fused output is needed when two or more test results are provided. Derived from Stahlschmidt et al. [54] and Responsible IT.	A combined score or estimate is returned when multiple tests are provided.
F3	The system produces an individual output for each provided test.	Per-test outputs help users inspect which tests support or weaken the combined result. Derived from Responsible IT and Amershi et al. [21].	Each provided test returns its own similarity score or composition estimate.
F4	The system handles missing tests in the combined output.	Users may provide only some tests. Derived from Responsible IT and Stahlschmidt et al. [54].	Available tests are combined, while missing tests are excluded from fusion.
F5	The interface presents the overall output and its contributing test outputs.	Users need to see how the combined result relates to the separate test outputs. Derived from Guideline 11 from Amershi et al. [21].	The combined output and contributing test outputs are available to the user.
F6	The interface supports QR-code reference selection.	Responsible IT identified QR-code selection as suitable for retrieving the intended reference fabric. Derived from Responsible IT and Guideline 4 from Amershi et al. [21].	The wireframe shows reference selection and entry of matching test results.
E1	The system frames outputs as recommendations and leaves the final decision to the user.	This avoids pass/fail verdicts and supports human oversight. Derived from Responsible IT, the EU AI Act [4], the People + AI Guidebook [10], and Guideline 2 from Amershi et al. [21].	Results are phrased as recommendations rather than definitive verdicts.
E2	The interface warns the user when outputs conflict, similarity falls below a configurable minimum, or no matching stored fabrics are found.	Users need uncertainty and conflict cues to calibrate trust. Derived from Responsible IT, the EU AI Act [4], the People + AI Guidebook [10], and Guidelines 2, 9, and 11 from Amershi et al. [21].	The warning identifies the relevant tests and suggests next steps, such as retesting or manual review.
L1	Submitted test data remains attributable to its owner and removable on request.	Companies may submit proprietary fabric data and must retain control over its use. Derived from GDPR [6] and Responsible IT.	Submitted test records include ownership metadata and the interface supports removing stored results.
O1	The system lets each organisation define test weights through a Best-Worst Method workflow.	Organisations may prefer certain tests depending on their context. Derived from Responsible IT and Rezaei [44].	The system applies the organisation-specific BWM-derived weights.
T1	A CNN produces the image-based test representation.	CNN embeddings provide a compact representation for image-based comparison. Derived from Responsible IT and Barros et al. [27].	CNN feature embeddings are extracted and used for image-based similarity scores.

Table 2. Requirements.

ID	Requirement	Justification	Definition of Done
T2	The composition-estimation output is evaluated using composition error.	The estimate is useful only if it is close to the fabric being characterised. Derived from Responsible IT and Kaya and Bilge [31].	Composition error is reported for the implemented training-mean baseline, image-only, image-only (softmax-weighted), spectra-only, image + spectra, and image (softmax-weighted) + spectra configurations.
T3	The system stores and retrieves test representations using a vector database.	Responsible IT requested PGVector, which supports vector storage and similarity search in PostgreSQL [11].	Encoded test results are stored as vectors in PGVector and retrieved for similarity-based comparison.

3.2 Concept

Before use, the organisation configures test weights. Equal weights are the default. For custom weights, a domain expert completes the BWM setup by selecting the most and least important tests for the use context. The expert then compares the remaining tests against these two reference points, after which the system derives BWM-based weights [44]. Once configured, the interface supports reference comparison and composition estimation. In both modes, the user can inspect the overall output, per-test outputs, similarity scores, warnings, and suggested next steps.

3.2.1 Iteration 1: Low-Fidelity Composition-Only Design.

The first low-fidelity design supported composition estimation only: users entered available test results, received an estimated composition, and could inspect separate test outputs, shown in Figure 2.

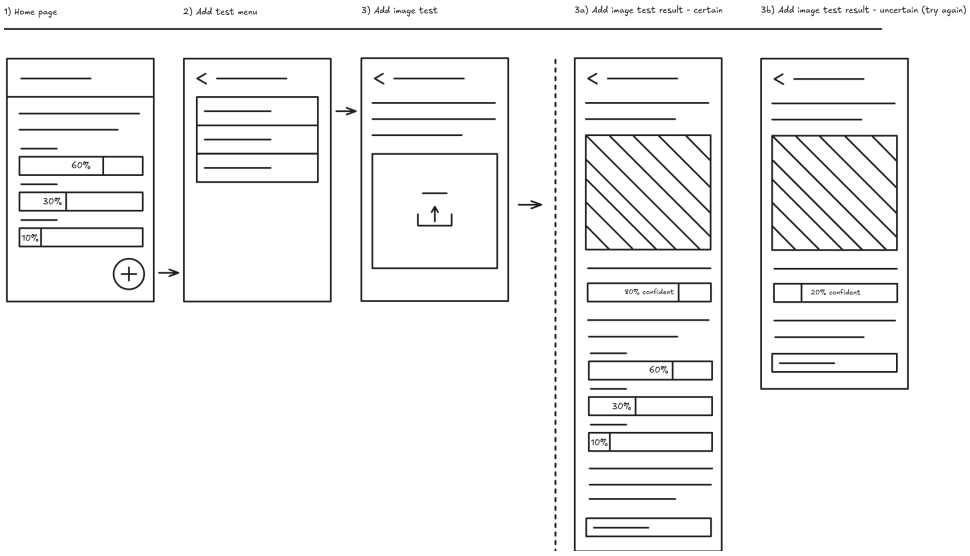


Fig. 2. First low-fidelity wireframe for composition estimation.

3.2.2 Iteration 2: Low-Fidelity Composition and Reference Design.

After Responsible IT feedback, the second low-fidelity design separated reference comparison from composition estimation. This distinguishes users with a stored reference fabric from users estimating composition from similar stored fabrics. Figure 3 shows the revision.

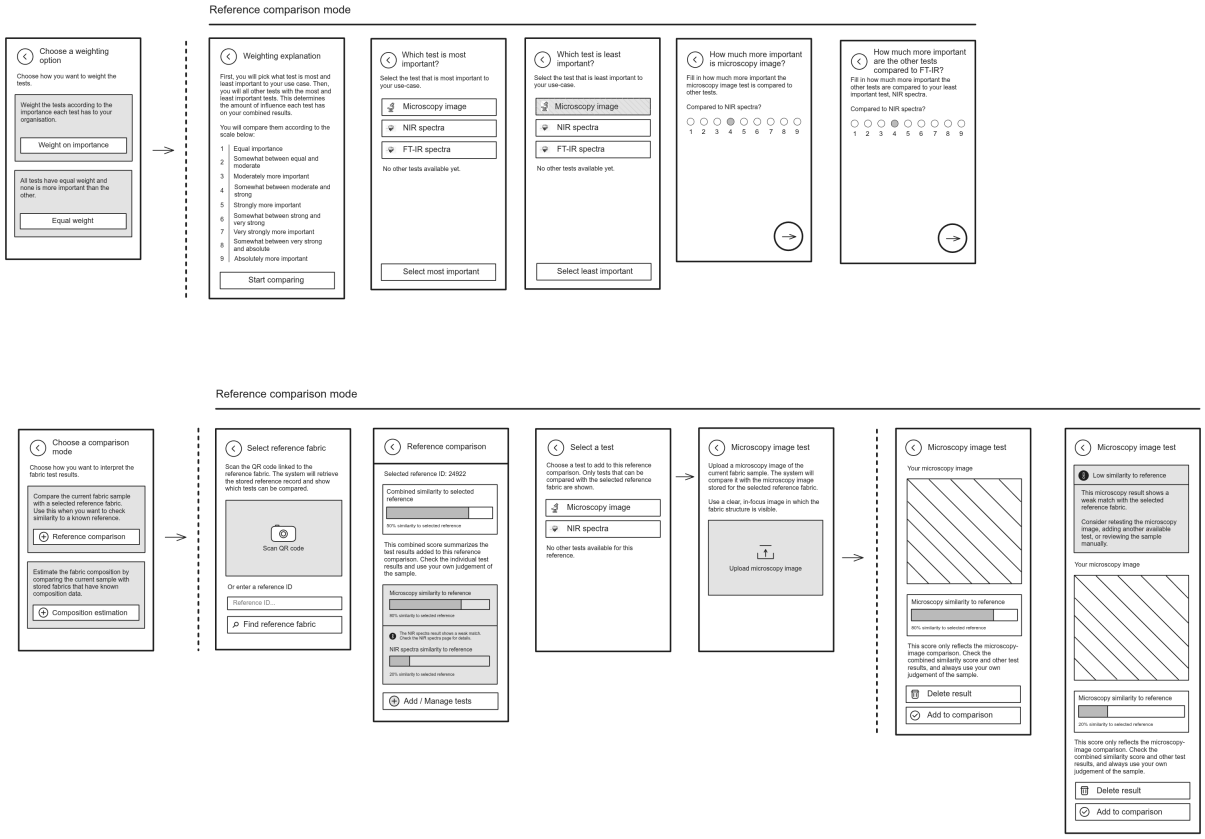


Fig. 3. Second low-fidelity wireframe with BWM weighting, mode selection, and reference-comparison flow.

3.2.3 *Happy Flows.*

Appendix A shows the composition-estimation happy flow, where the user enters available test results, receives an estimated composition, and can inspect per-test results. Appendix B shows the reference-comparison happy flow, where the user scans the reference QR code, enters matching test results, and receives per-test and combined similarity scores.

3.2.4 *Error Flows.*

Appendix C shows the error flows for missing references, unavailable comparable test data, low similarity, and conflicting test outputs. These flows show when the user can choose another reference, switch to composition estimation, exclude weak test outputs, or review a warning before continuing.

3.2.5 *AI Breakdown.*

Table 3 summarises the AI breakdown. Low-impact mistakes interrupt the workflow without producing a misleading material interpretation. Medium-impact mistakes reduce the available evidence. High-impact mistakes can produce an incorrect material interpretation if accepted without review.

Table 3. AI breakdown.

In	Task	Out	Possible mistakes	Impact of mistakes	Mitigations
Microscopy image or NIR spectrum	Represent the test result and compare it with stored representations	Similarity scores and retrieved fabrics with fibre-composition metadata	The representation retrieves fabrics with different compositions	High	Show the result as a separate test output, warn when similarity is below the configurable minimum or other tests disagree, and suggest retesting, adding another test, or manual review
Per-test outputs	Check whether tests agree before presenting the combined output	Combined similarity score or composition estimate	Tests point to different fibre labels or strongly different similarity scores	High	Explain that the tests are not aligned, keep per-test outputs visible, suggest retesting or adding another test, and avoid a definitive verdict
Entered test result	Search stored fabrics with the required test data	Retrieved stored fabrics for the provided test	No stored fabrics are available for the required test data	Medium	Explain that no matching stored fabrics were found and suggest adding another test, retesting, or manual review
Per-test output	Decide whether the output can contribute to the combined result	Included or excluded test output	The similarity score is below the configurable minimum for that test	Medium	Exclude the test from the combined result, keep the separate output visible, and suggest retesting
Reference QR code	Retrieve and check the selected reference	Stored reference fabric with matching test data	The reference cannot be found or has no matching saved test data	Low	Explain the issue and let the user try again, choose another reference, or switch to composition estimation

3.2.6 *Level of Automation.*

Kore describes ten levels of autonomy, where higher levels give AI more control and require more user trust [34]. The prototype is positioned in the augmentation range, closest to level 4. It suggests an interpretation by producing a similarity score or estimated composition, but it does not make or execute a decision. The user remains in the loop and decides whether to accept the result, question it, retest, add another test, or choose manual review. This reduces the risk that an incorrect output is automatically turned into a validation, sorting, or product-information decision affecting recyclers, sellers, or consumers. The same approach could later support more automated use cases, for example by suggesting next actions while keeping the underlying test evidence visible to the user.

3.3 Data

Table 4 compares textile datasets against the need for fabric-level fibre-content metadata and more than one test type per sample. TextileNet [58], CottonFabricImageBD [38], and the OCT textile-fabric dataset [46] are image datasets without multimodal fabric-level measurements. OpenTextile-NIR [53] includes NIR data and fibre-composition metadata but only 71 textile samples for image-model training. NIR-SORT [9] is selected because it contains fabric samples with spectroscopy, microscopy, and fibre-content composition metadata.

Table 4. Comparison of textile datasets.

Dataset	Textile level	Data types	Size	Provenance of labels	Main limitation
TextileNet [58]	Fashion and garment-oriented textile images	Fashion RGB images with fibre and fabric taxonomy labels	760,949 images	Taxonomy labels generated with material scientists	Image-only; no fabric-level multimodal measurements
CottonFabricImageBD [38]	Cotton-blend fabric images	RGB images of coloured cotton fabrics	1,300 original and 27,300 augmented images across 13 cotton-percentage classes	Cotton percentages from fabric stores and manufacturers	Image-only; limited to cotton-percentage classes
OpenTextile-NIR [53]	Post-industrial textile samples	NIR hyperspectral images, RGB photographs, representative spectra, and metadata	71 samples, over 11 million spectra, and over 6 million annotated spectra	Supplier-provided fibre-composition metadata	Few unique samples for image-model training
OCT textile fabrics [46]	Single-material fabric samples	Optical coherence tomography images	9 fabrics, with at least 100 scans per fabric surface	Pure-material labels confirmed through fire testing and microscopy	OCT-only; limited to three single-material classes
NIR-SORT [9]	Known-provenance, pre-consumer, and post-consumer fabric samples	Benchtop and handheld NIR spectra, optical microscopy images, and fibre-content metadata	113 fabrics, approximately 4,248 spectroscopy files, and approximately 1,044 microscopy files	Fibre-content and provenance metadata	Limited fabric count for image-model training

NIR-SORT contains known-provenance, pre-consumer, and post-consumer specimens: Testfabrics, Inc. samples, local secondhand garments, and unused fabric samples. It includes manufacturer-provided and garment-derived metadata. NIR spectroscopy was recorded under ambient laboratory conditions on non-overlapping yarn regions, and microscopy images were captured from flat, single-layer fabric samples using a microscope-mounted camera [9].

3.3.1 *Data Splitting.*

Although NIR-SORT is documented as 113 fabrics [9], the downloaded files contain 155 distinct sample IDs linked to measurement files and fibre-composition metadata. These IDs are therefore used as split units, with all measurements from one fabric assigned to one split. Microscopy samples have 4 to 12 images each, with a median of 6. Spectroscopy records have 2 to 127 files each, with a median of 42. Before splitting, CNN eligibility is determined at fabric-sample level using Table 5, and groups are used for stratification.

Table 5. CNN-eligibility decisions for the fabric-sample split.

Decision	Reason	Result
Use only fabrics with a dominant fibre above 50% for CNN classification.	No-dominant and tied compositions would create too many small CNN classes.	143 fabric samples are CNN-eligible; 3 no-dominant and 8 tied-dominant samples are excluded from CNN classification.
Keep only CNN classes with at least three fabric samples.	Each class needs at least one fabric sample in training, validation, and testing.	Acrylic has one fabric sample and is excluded.

The split groups are Cotton (76), Polyester / recycled polyester / PET (44), Synthetic polyamide / nylon (13), EXCLUDED_CNN (12), Regenerated cellulosic (6), and Natural polyamide (4). EXCLUDED_CNN remains available for retrieval and composition estimation but is excluded from CNN training and classification metrics. Stratified splitting uses a 70%/15%/15% train, validation, and test ratio. Validation is used for model selection and parameter choices. The locked test split is used only for final evaluation. Table 6 summarises the split result.

Table 6. Fabric-sample and measurement-file counts per data split.

Split	Fabric samples	Image measurement files	Spectroscopy measurement files
Training	107	729	3101
Validation	24	149	530
Test	24	162	607

Figure 4 shows the resulting training split distribution.

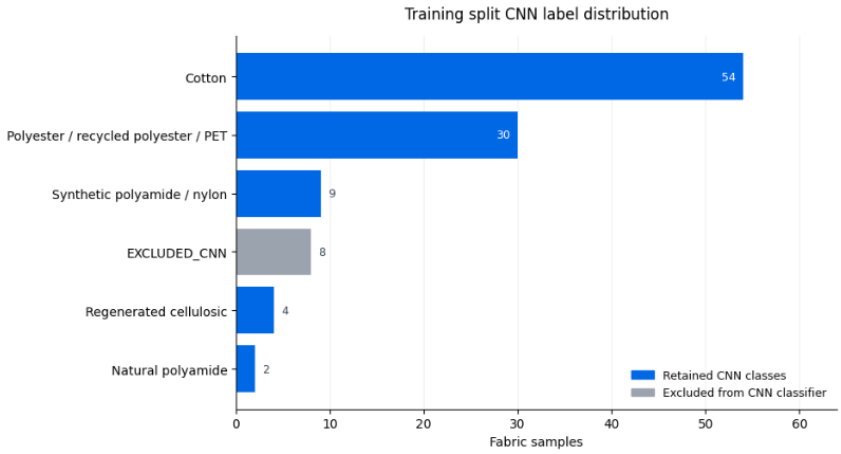


Fig. 4. Training split distribution by CNN split group.

3.3.2 Image Preprocessing.

For the image model, colour, mono, and polarized microscopy images from CNN-eligible fabrics are converted to grayscale to reduce colour dependence and focus on texture. Polarized images are kept because they are valid microscopy measurements and preserve image variation. The CNN image set contains 99 training fabrics, 22 validation fabrics, and 22 test fabrics, with 665, 139, and 152 microscopy images respectively. During training, each selected training image receives one augmented copy to increase visual variation [50]. Validation and test images are not augmented. The augmentation values are kept small to preserve fabric identity: rotation up to 10°, horizontal and vertical flips with 50% probability, and random contrast adjustment up to 10%. Rotations and flips represent orientation changes, while contrast adjustment represents lighting differences.

3.3.3 Spectral Preprocessing.

The selected NIR input is Benchtop N/N-A, a benchtop absorbance representation covering all 155 fabric samples. It contains 717 spectroscopy files, with 1 to 7 files per fabric and a median of 7. Benchtop N/N-R also covers all samples, but is not evaluated separately because it is the reflectance version of the same benchtop measurements. Each spectrum is stored as 4149 ordered absorbance values, where matching positions correspond to the same wavelengths, and is compared using cosine similarity.

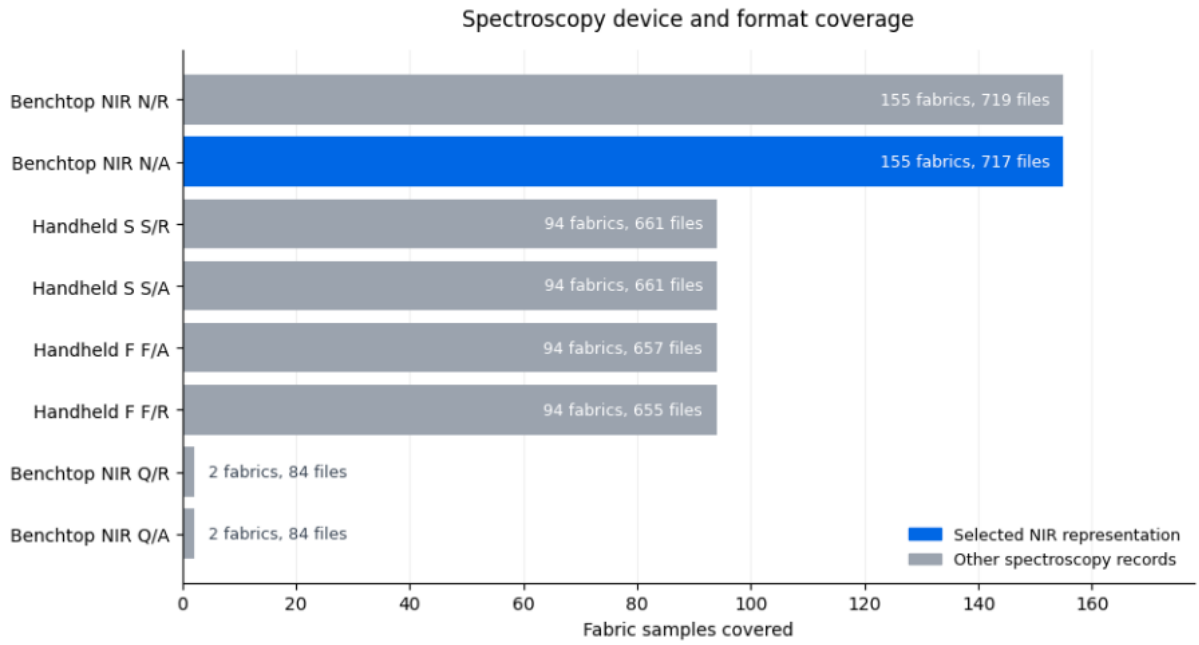


Fig. 5. Fabric coverage for the available spectroscopy inputs.

For retrieval, spectra from the same fabric are averaged before vector storage or lookup. Each training fabric contributes one stored spectra vector, and each validation or test fabric contributes one query vector, keeping retrieval fabric-level and preventing fabrics with more spectra files from contributing more records.

3.4 Model & Architecture

This section describes the late-fusion structure, output fusion, and CNN image model.

3.4.1 Late Fusion Architecture.

The architecture uses late fusion because the available tests can differ between fabrics and users. Early or intermediate fusion would require a fixed input set, or additional mechanisms for tests that are missing. Here, each test is processed on its own first. The outputs are only combined after each available test has produced a test-specific result.

For retrieval, each test output is encoded as a vector and stored in PGVector, a PostgreSQL vector database requested by Responsible IT [11, 17]. Stored fabrics are retrieved per test type.

Cosine similarity is used for vector comparison. Unlike Euclidean or Manhattan distance, which can be affected by magnitude and scale differences across variables [16], cosine similarity measures vector angle and is commonly used for high-dimensional vector-database representations [18]. Calculating similarity per test also prevents high-dimensional image embeddings from overshadowing spectra vectors and supports missing tests, since only overlapping available tests are compared. It also makes the architecture easier to extend: a new test type can be added by giving it its own representation and adding its output to the final fusion step, instead of redesigning one model that expects all inputs. This supports architecture-level scalability in line with CRISP-ML(Q) [55].

3.4.2 Weighting and Output Fusion.

The system has two interpretation modes. Reference comparison produces a combined similarity score against a selected reference fabric. Composition estimation estimates fibre composition from retrieved stored fabrics. In both modes, each available test first produces its own output. The BWM weight w_t represents the organisation's importance assigned to test t. A confidence value c_t may indicate saved-output reliability, but it affects fusion only when every included output has such a value. Otherwise $c_t = 1$.

In reference comparison, only tests available for both the characterised fabric and selected reference contribute. Each included test produces similarity s_t, combined with BWM weights and, where available for every included output, confidence values:

$$S = \frac{\sum_{t \in T} w_t c_t s_t}{\sum_{t \in T} w_t c_t}$$

Here, T is the set of included tests, w_t the BWM weight, c_t the confidence value, and s_t the test-specific similarity score. Confidence is applied only when every included test output provides it. Otherwise all c_t values remain 1. For example, if image and spectra weights are both 0.5 and their similarities are 0.6 and 0.9, the combined score is 0.75 with neutral confidence.

In composition estimation, each available test retrieves up to three comparable stored neighbours. The value $k = 3$ is an untuned prototype setting: it keeps estimates focused on a few similar fabrics, but should be tuned per test type in future work. A configurable minimum similarity can exclude low-similarity neighbours.

Retrieved fabrics have labelled fibre-composition fractions. Before averaging, fibre labels are aligned and missing fibres are treated as 0. For example, averaging 0.8 cotton / 0.2 polyester with 0.7 cotton / 0.3 elastane gives 0.75 cotton, 0.10 polyester, and 0.15 elastane. More generally, selected neighbours for test t are denoted K_t, with C_t as the test-level composition estimate:

$$C_t = \frac{\sum_{r \in K_t} c_r C_r}{\sum_{r \in K_t} c_r}$$

Here, C_r is the labelled composition of retrieved neighbour r, and c_r is the neighbour confidence value. By default, $c_r = 1$. For images, the highest CNN softmax output is used as an uncalibrated confidence proxy. Spectra confidence is unavailable and remains 1. Per-test composition estimates are then fused with BWM weights:

$$\hat{C} = \frac{\sum_{t \in T} w_t C_t}{\sum_{t \in T} w_t}$$

Here, T is the set of available tests that produce a test-level composition estimate C_t. A test may be excluded when missing, when no stored neighbours are available, or when all retrieved neighbours fall below the configurable minimum similarity.

3.4.3 *CNN Development.*

The CNN uses DenseNet201 pretrained on ImageNet [28]. Pretraining is useful for the small microscopy dataset because learned visual features can be reused, but it also creates a domain gap because ImageNet contains natural images rather than textile microscopy images. Table 1 compares earlier textile-image classification approaches. DenseNet201 is selected based on Barros et al. [27], who reported the strongest result for DenseNet201. Their result supports the architecture choice, but is not a direct baseline because it used mobile-camera fabric images rather than NIR-SORT microscopy images.

The CNN is trained through classification-based transfer learning. It predicts the dominant fibre class, while the same 1920-dimensional embeddings are used for image retrieval. Contrastive learning would optimise retrieval embeddings more directly, but adds training complexity [33]. The first iteration therefore keeps DenseNet201 frozen to test whether pretrained embeddings are already useful for image-based tasks. The second iteration fine-tunes the final DenseNet201 feature block and classifier head to test whether limited adaptation improves those embeddings, following transfer-learning work on limited-data visual tasks [40].

To align with CRISP-ML(Q), model complexity is evaluated alongside performance instead of treating the best score as the only quality criterion [55]. The baselines provide the simplest reference points: the majority-class baseline for CNN classification and the training-mean baseline for composition estimation. The first CNN iteration then adds a frozen DenseNet201 backbone and trains only a linear head. The second iteration is more complex because it also updates the final DenseNet201 feature block, while earlier layers remain frozen to reduce overfitting risk on the small dataset. This orders the evaluation from simple baselines to a more adapted image model and shows whether any improvement justifies the added training complexity.

Figure 6 summarises the second-iteration CNN path: the DenseNet201 features produce both the 1920-dimensional image embedding for retrieval and the five class outputs for CNN classification.

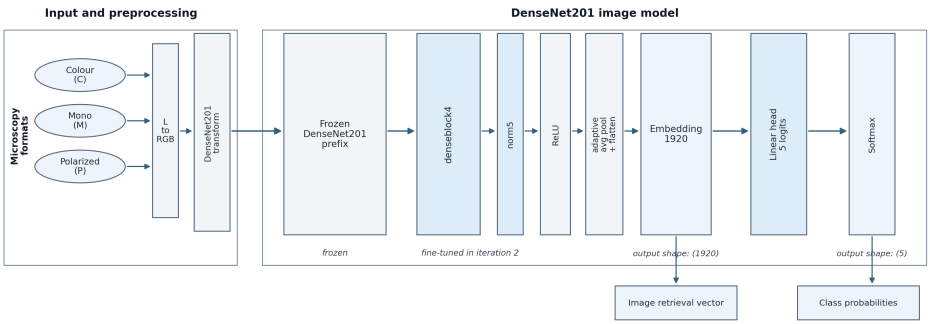


Fig. 6. CNN image architecture, showing the second-iteration trainable modules.

3.4.4 *First CNN Iteration.*

In the first iteration, DenseNet201 is frozen and only the classification head is trained. This provides a frozen-backbone baseline for checking whether pretrained embeddings are useful before fine-tuning. Table 7 lists the training settings.

Table 7. Parameters for the first CNN iteration.

Parameter	**Value**	**Rationale**
Backbone training	Frozen DenseNet201	Tests pretrained DenseNet201 features before fine-tuning.
DenseNet201 feature vector size	1920	Output size of the selected DenseNet201 feature extractor.
Classification head	Linear layer	Tests whether the frozen embeddings are linearly separable for the retained CNN classes.
Output units	5	Matches the five retained CNN classes.
Loss function	Class-weighted cross-entropy	Handles class imbalance by increasing the contribution of minority classes.
Optimizer	AdamW	Standard, well-established optimizer for CNN training [37].
Learning rate	1×10^{-3}	Initial learning rate for the classification head.
Weight decay	1×10^{-4}	Small regularisation value to reduce overfitting [37].
Batch size	64	Batch size that balances gradient stability and memory usage.
Maximum epochs	80	Upper training limit used with early stopping.
Early stopping patience	12 epochs	Stops training when validation performance does not improve for several epochs.
Random seed	42	Fixed seed for reproducibility.

3.4.5 *Second CNN Iteration.*

The second iteration starts from the first-iteration checkpoint and fine-tunes `denseblock4`, `norm5`, and the classifier head. Earlier DenseNet201 layers remain frozen, focusing the change on the final image feature block. Table 8 lists the updated settings.

Table 8. Changed and added parameters for the second CNN iteration.

Parameter	**Value**	**Rationale**
Initialisation	First-iteration checkpoint	Starts from the trained frozen-backbone classifier.
Trainable backbone modules	`features.denseblock4`, `features.norm5`	Unfreezes the last DenseNet201 feature block while keeping earlier generic features frozen.
Backbone learning rate	1×10^{-4}	Updates pretrained DenseNet201 weights more slowly than the head to help preserve learned features.

3.5 Evaluation

The evaluation covers both the individual model components and the final characterisation output. Following CRISP-ML(Q), it assesses predictive performance together with explainability, scalability, model complexity, and representation quality [55]. For the CNN classifier, this includes fabric-level weighted-F1, accuracy, and a confusion matrix, complemented by Grad-CAM visualisations that show which image regions influence predictions and a timing experiment that measures inference scalability. Image and spectra representations are evaluated through cosine-similarity checks, and the final characterisation output is assessed through composition-estimation error across all tested configurations. Validation results guide model selection and parameter choices; the locked test split is used only for final evaluation.

3.5.1 *Evaluation of the CNN Classifier.*

Classification models are commonly evaluated with accuracy, precision, recall, and F1-score [52]. For the CNN, the primary metric is fabric-level weighted-F1: because each fabric has multiple microscopy images, class probabilities are first averaged across a fabric's images, and the class with the highest average becomes the fabric-level prediction. Weighted-F1 is then computed from these predictions, weighting each class by its number of validation fabrics to reflect the class distribution and prevent any single over-represented fabric from distorting the score. A confusion matrix is reported to inspect performance on the smaller classes separately, and accuracy is included as a secondary descriptive metric. The CNN is also compared against a majority-class baseline.

3.5.2 *Evaluation of CNN Inference Scalability.*

CNN inference scalability is evaluated for the second CNN iteration. Validation images are preprocessed once and repeated deterministically into workloads of 8, 16, 32, 64, 128, 256, 512, 1024, and 2048 images. Each workload uses inference batch size 64, five timing repeats, and one warm-up run. Timing includes the DenseNet201 prefix and fine-tuned tail returning softmax probabilities and embeddings, showing how inference scales after preprocessing.

3.5.3 *CNN Heatmap Visualisation.*

Grad-CAM supports explainability by highlighting image regions that contribute to CNN predictions [49]. Nine correctly classified validation images are selected from the three largest CNN classes: cotton, polyester / recycled polyester / PET, and synthetic polyamide / nylon. Within each class, images are ranked by second-iteration softmax confidence, and the lowest-, median-, and highest-confidence examples are visualised for both CNN iterations. For final test evaluation, the same selection logic is applied to correctly predicted test images from the second iteration only.

3.5.4 *Evaluation of Test-Representation Similarity.*

The similarity check tests whether related measurements are closer than unrelated measurements in image and spectra representation spaces, following the contrastive-learning principle that related samples should be closer and unrelated samples further apart [33]. Pairs are grouped as same-fabric, same dominant class but different fabric, and different dominant class. For images, each item is a microscopy image embedding. For spectra, each item is a Benchtop N/N-A spectra measurement. Cosine similarity is used for both.

The validation split is used to compare the frozen and fine-tuned image representations. Because the spectra representation does not change, the comparison mainly shows how fine-tuning affects the image embedding space. The same check is repeated on the test split for final evaluation.

Because labels are dominant fibre-composition labels rather than pure materials, fabrics within one class can contain different blends and fabrics from different classes can share fibres. The aim is useful relative similarity structure, not perfect separation.

3.5.5 *Evaluation of the Combined Characterisation.*

Composition estimation is evaluated for the training-mean baseline, image-only, image-only (softmax-weighted), spectra-only, image + spectra, and image (softmax-weighted) + spectra configurations. In both image and spectra retrieval, $k = 3$ similar stored fabrics are retrieved using CNN image embeddings or averaged fabric-level spectra vectors. The labelled compositions of the retrieved fabrics are then averaged. In the softmax-weighted image configurations, retrieved image neighbours contribute more to the average when the CNN assigned them a higher softmax score. This score is used only as an uncalibrated confidence proxy: it indicates how strongly the CNN treats the image as belonging to its predicted class. Combined configurations fuse image-based and spectra-based estimates using Section 3.4.2 with default equal test weights. The training-mean baseline ignores input tests and predicts the average labelled composition of the training fabrics.

All configurations use mean absolute composition error [8]. For evaluated fabric sample j, with estimated composition $\hat{C}_j$, the composition error is:

$$E_j = \frac{1}{2} \sum_{f \in F} |C_{j,f} - \hat{C}_{j,f}|$$

The mean absolute composition error is then:

$$MAE_{comp} = \frac{1}{n} \sum_{j=1}^{n} E_j$$

Here, n is the number of evaluated fabric samples, F the aligned set of fibre labels, $C_{j,f}$ the labelled value for fibre f, and $\hat{C}_{j,f}$ the estimated value. Missing fibres are treated as 0. The factor $\frac{1}{2}$ prevents one composition shift from being counted twice.

4 Results

4.1 First Iteration

The first CNN iteration was evaluated per fabric on the validation split: predictions from 139 microscopy images were averaged into 22 validation-fabric predictions. Table 9 compares the frozen DenseNet201 linear head with the majority-class baseline. Its higher weighted-F1 score suggests that pretrained image embeddings are useful for classification without fine-tuning.

Table 9. CNN validation results for the first iteration.

Model	**Weighted-F1**	**Accuracy**
Frozen DenseNet201 linear head	0.861	0.864
Majority-class baseline	0.333	0.500

Figure 7 shows that cotton and polyester / recycled polyester / PET, the most common classes, are predicted correctly in most cases. The smaller classes, natural polyamide, regenerated cellulosic, and synthetic polyamide, contain too few validation fabrics for stable class-level conclusions.

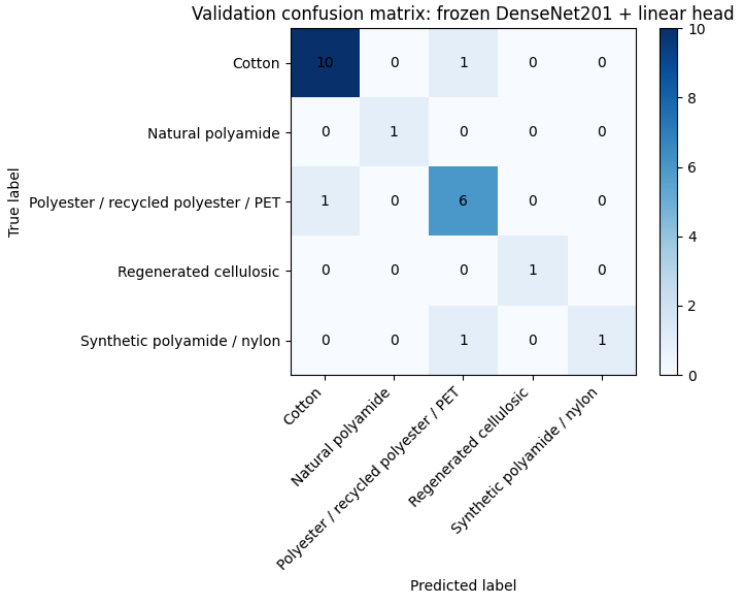


Fig. 7. Validation confusion matrix for the frozen DenseNet201 linear head.

4.1.1 Composition Estimation.

Composition estimation used the 22 CNN-eligible validation fabrics as queries and 99 training fabrics as the stored reference set. Table 10 reports the validation composition error for each configuration. The image + spectra configuration uses unweighted image and spectra estimates. The softmax-weighted image configurations weight each retrieved image neighbour by its highest softmax output from the frozen DenseNet201 linear head.

Table 10. Validation composition-estimation results for the first iteration.

Configuration	**MAE_{comp}**
Training-mean baseline	0.518
Image only	0.241
Image only (softmax-weighted)	0.236
Spectra only	0.113
Image + spectra	0.175
Image (softmax-weighted) + spectra	0.172

Spectra-only retrieval has the lowest validation composition error, while image-only retrieval has the highest error among retrieval-based configurations. Image-only retrieval still performs better than the training-mean baseline, so the frozen DenseNet201 embeddings contain some composition-related information, but less than the spectra representation. The combined image + spectra results fall between the two individual test types. Softmax weighting slightly reduces the image-only error from 0.241 to 0.236 and the combined error from 0.175 to 0.172, but the gap with spectra-only retrieval remains. The second iteration therefore targets the image representation.

4.2 Second Iteration

The second CNN iteration uses the same validation split, again averaging predictions from 139 microscopy images across 22 validation fabrics. It fine-tunes the final DenseNet201 feature block together with the five-output linear head. Table 11 compares the majority-class baseline, the frozen first-iteration model, and the fine-tuned second-iteration model.

Table 11. CNN validation results for the second iteration.

Model	**Weighted-F1**	**Accuracy**
Majority-class baseline	0.333	0.500
Frozen DenseNet201 linear head	0.861	0.864
Fine-tuned DenseNet201 final feature block	0.906	0.909

Fine-tuning improves validation weighted-F1 from 0.861 to 0.906 and accuracy from 0.864 to 0.909. In absolute terms, this is one additional correctly classified validation fabric. Figure 8 shows 20 of the 22 validation fabrics classified correctly. One cotton fabric and one synthetic polyamide / nylon fabric are predicted as polyester / recycled polyester / PET.

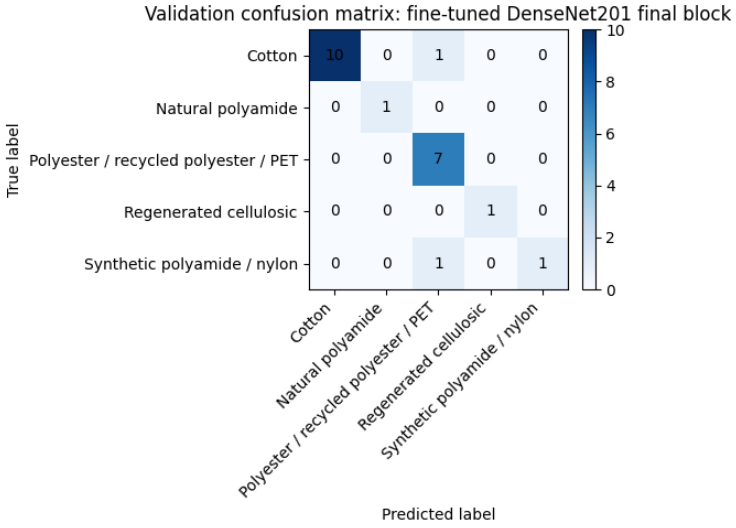


Fig. 8. Validation confusion matrix for the fine-tuned DenseNet201 final feature block.

4.2.1 Composition Estimation.

Composition estimation was repeated with the fine-tuned image embeddings and the same validation and stored reference sets. Table 12 reports the validation composition error for each configuration.

Table 12. Validation composition-estimation results for the second iteration.

Configuration	**MAE_{comp}**
Training-mean baseline	0.518
Image only	0.221
Image only (softmax-weighted)	0.221
Spectra only	0.113
Image + spectra	0.164
Image (softmax-weighted) + spectra	0.164

Fine-tuning reduces the image-only validation composition error from 0.241 to 0.221 and the image + spectra error from 0.175 to 0.164. Spectra-only retrieval remains strongest at 0.113, and image softmax weighting has no visible effect after rounding to three decimals. The gap between spectra-only and image-only retrieval supports keeping separate test outputs visible next to the combined estimate.

4.2.2 Grad-CAM Heatmaps.

Figure 9 compares Grad-CAM heatmaps for selected microscopy images in both CNN iterations. The first iteration generally produces diffuse heatmaps, with highlighted regions spread over larger image areas. After fine-tuning, the heatmaps become more localised. In several examples, especially for polyester / recycled polyester / PET, the highlighted regions align with visible fabric structures. This suggests that fine-tuning changes how the model uses image regions compared with the frozen model.

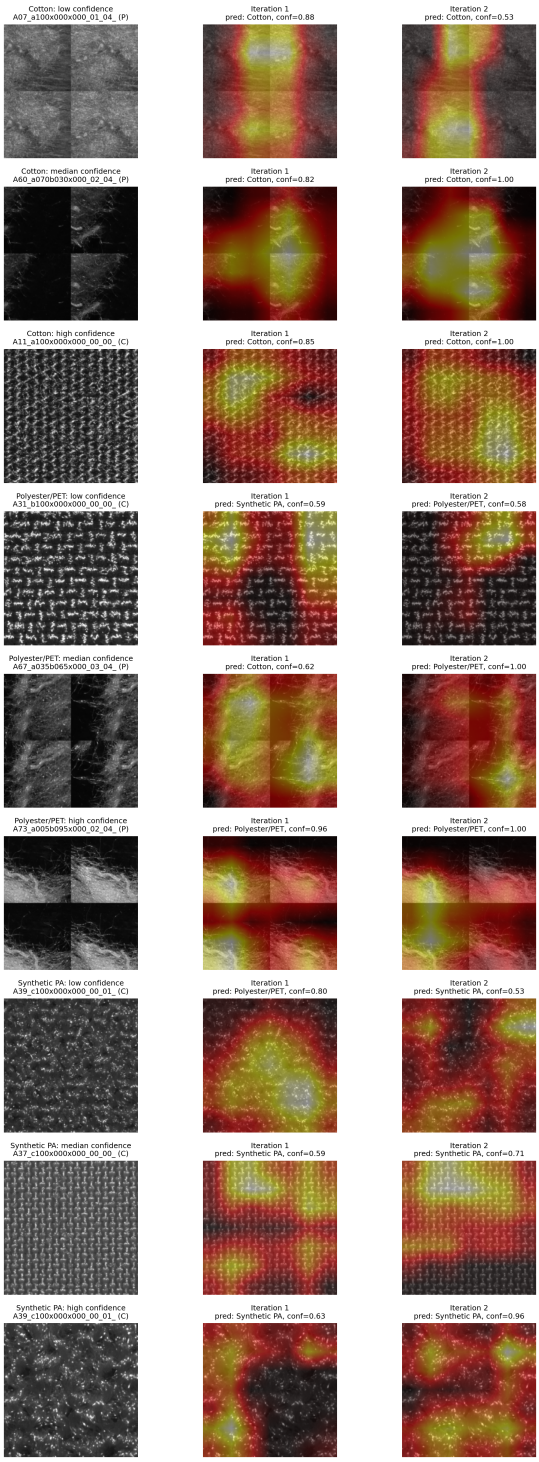


Fig. 9. Grad-CAM heatmaps for selected validation microscopy images in the first and second CNN iterations.

4.2.3 Test-Representation Similarity.

Figure 10 shows validation image-embedding similarities for the frozen and fine-tuned CNN representations. The frozen representation has higher cosine similarities overall. In both representations, same-fabric pairs have the highest median similarity and are partly separated from the other groups, although the distributions still overlap. In the frozen representation, same-class different-fabric pairs and different-class pairs have similar distributions and medians, suggesting little separation between dominant class groups. After fine-tuning, similarity values shift lower overall. Same-fabric pairs remain highest, and same-class different-fabric pairs have a slightly higher median than different-class pairs, but the groups still overlap substantially.

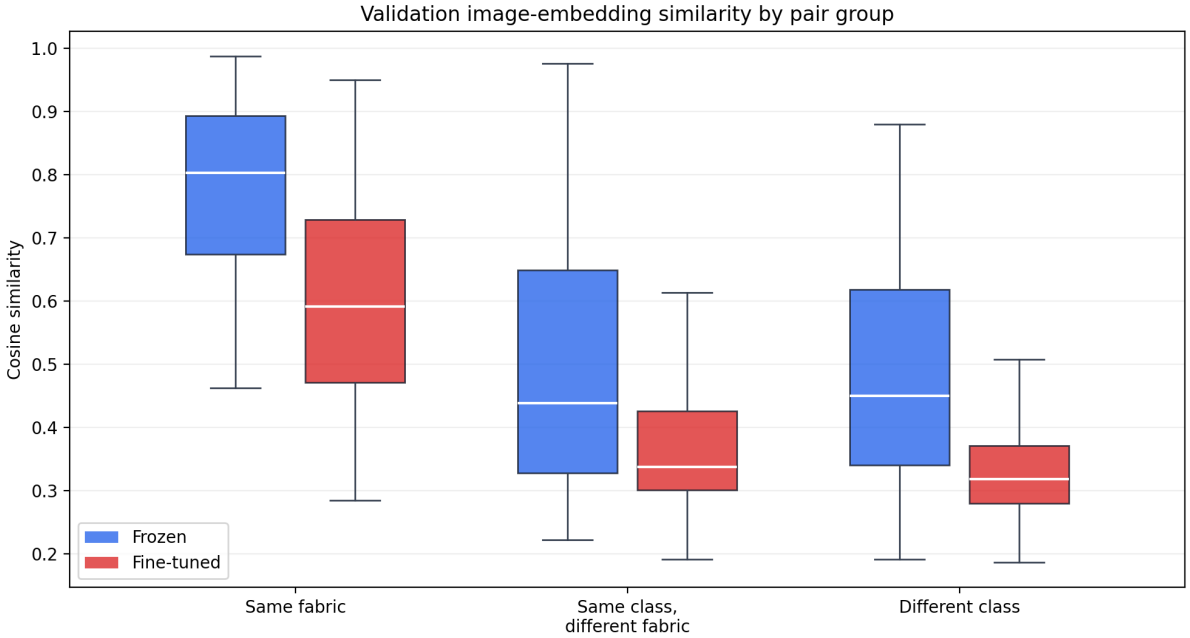


Fig. 10. Validation image-embedding similarity by pair group.

Figure 11 shows validation spectra similarities by pair group. Same-fabric spectra pairs are concentrated at very high cosine similarities, with all pairs above 0.99. The other two groups have wider distributions, but their medians follow the expected order: same-fabric pairs are highest, same-class different-fabric pairs are next, and different-class pairs are lowest. The latter two groups are not fully separated, but this ordering indicates useful relative similarity structure.

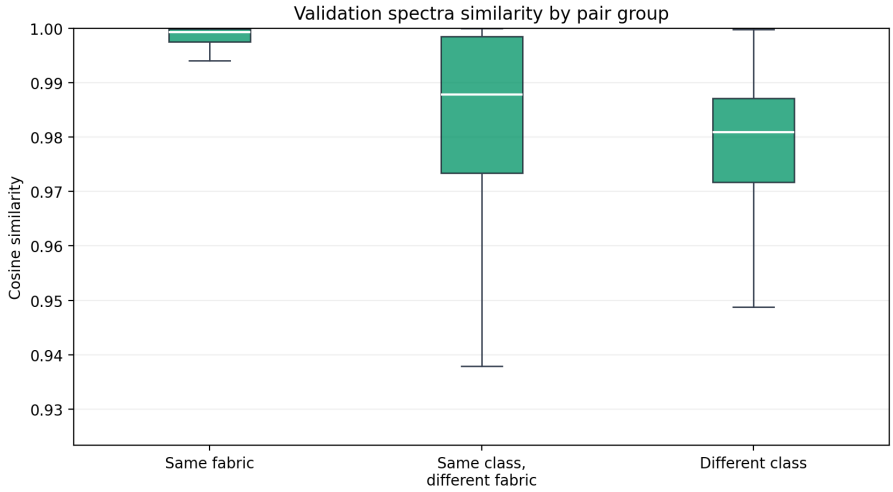


Fig. 11. Validation spectra similarity by pair group.

These differences affect how similarity-based outputs should be interpreted. Image and spectra representations both use cosine similarity, but image similarities cover a wider range while spectra similarities are concentrated near the top of the scale. A score should therefore be interpreted relative to the test representation that produced it. For composition estimation, one shared minimum similarity value is unsuitable for all test types: an image-selective threshold may be too low for spectra, while a spectra-suited threshold may exclude too many image neighbours. For reference comparison, the same cosine value can therefore provide different evidence depending on the test type.

4.2.4 CNN Inference Scalability.

Figure 12 shows inference timing for the second CNN iteration on repeated validation images. Each timed run passes the images through the DenseNet201 prefix and fine-tuned tail, producing both softmax probabilities and embeddings. The measurements were collected on an NVIDIA GeForce GTX 1060 6GB. Inference time increases approximately linearly with image count. From 64 images upward, throughput stays between 122 and 126 images per second. The average processing time is about 8.10 ms per image; the largest workload contains 2048 images and takes 16.83 seconds at 122 images per second.

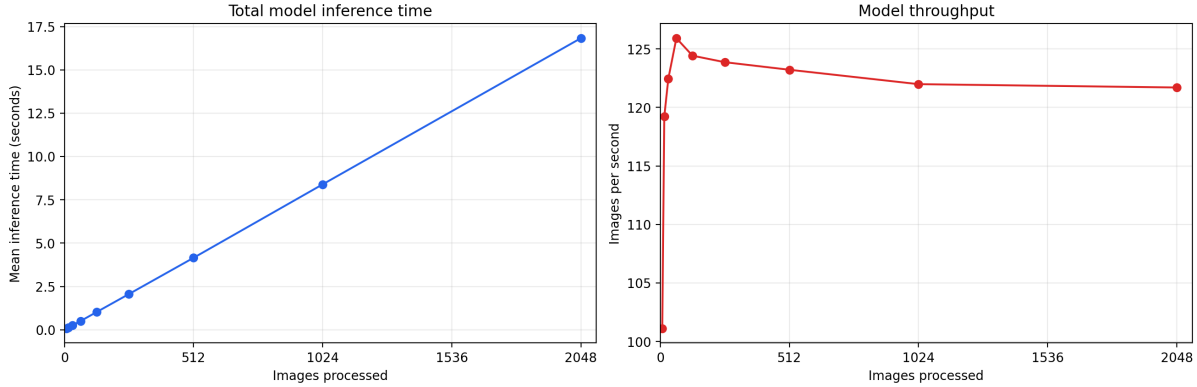


Fig. 12. CNN inference scalability for the second iteration, measured after preprocessing on repeated validation images.

4.3 Final Test Evaluation

The final evaluation uses the locked test split with the second CNN iteration. The test split contains 152 microscopy images across 22 CNN-eligible test fabrics. For composition estimation, the 99 training fabrics form the stored reference set, while the test fabrics are used only as queries and evaluation targets.

4.3.1 CNN Classification.

Table 13 reports the test-set CNN classification results for the fine-tuned DenseNet201 model. The model outperforms the majority-class baseline, but performs lower on test than on validation: weighted-F1 decreases from 0.906 to 0.750, and accuracy decreases from 0.909 to 0.773.

Table 13. CNN test results for the second iteration.

Model	**Weighted-F1**	**Accuracy**
Majority-class baseline	0.333	0.500
Fine-tuned DenseNet201 final feature block	0.750	0.773

Figure 13 shows that 17 of the 22 test fabrics are classified correctly. Cotton is predicted correctly for 10 of 11 test fabrics, and polyester / recycled polyester / PET for 5 of 7. The small classes contain only one or two test fabrics each, so their results represent individual cases rather than stable class-level performance estimates. The natural polyamide test fabric is classified correctly, while the regenerated cellulosic test fabric is predicted as cotton and one synthetic polyamide / nylon fabric is predicted as polyester / recycled polyester / PET.

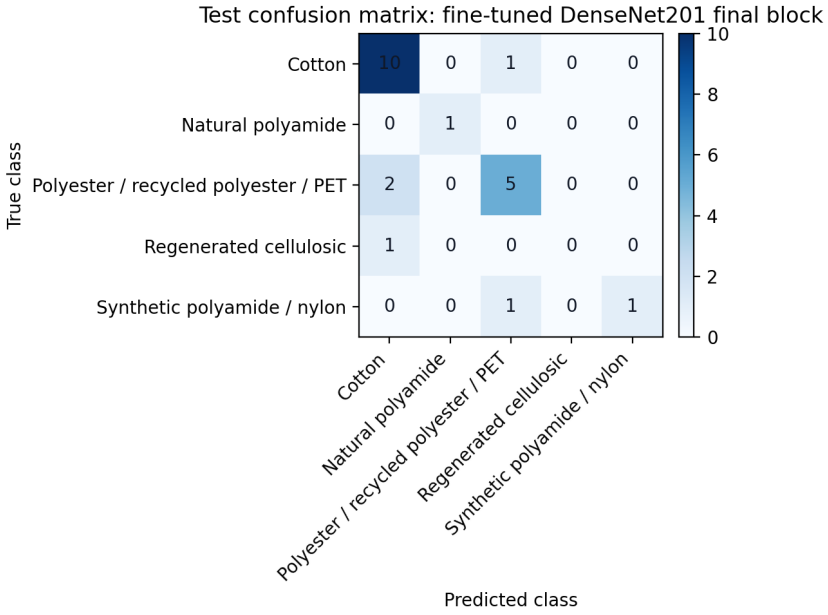


Fig. 13. Test confusion matrix for the fine-tuned DenseNet201 final feature block.

4.3.2 Composition Estimation.

Table 14 reports final test composition-estimation errors. Spectra-only retrieval remains strongest, with a test MAE_{comp} of 0.186. Image-only retrieval performs better than the training-mean baseline, but worse than spectra-only retrieval. The image + spectra result again lies between the two individual test types, showing that the combined estimate can hide modality-specific differences.

Table 14. Test composition-estimation results for the second iteration.

Configuration	**MAE_{comp}**
Training-mean baseline	0.484
Image only	0.230
Image only (softmax-weighted)	0.230
Spectra only	0.186
Image + spectra	0.204
Image (softmax-weighted) + spectra	0.204

Softmax weighting again has no visible effect after rounding to three decimals. Compared with validation, test errors are higher for the retrieval-based configurations, especially spectra-only retrieval, which increases from 0.113 to 0.186.

4.3.3 Test-Representation Similarity.

Figure 14 shows test-set representation similarities for the second-iteration image embeddings and spectra records. For image embeddings, same-fabric pairs have the highest median cosine similarity at 0.541, followed by

same-class different-fabric pairs at 0.338 and different-class pairs at 0.308. This ordering matches the intended use of the representation, although the groups overlap. Spectra records show the same ordering, but with values concentrated near 1.0: 0.999 for same-fabric pairs, 0.993 for same-class different-fabric pairs, and 0.988 for different-class pairs. This confirms the validation pattern that spectra similarities occupy a narrower and higher score range, so score ranges should be interpreted separately for each test type.

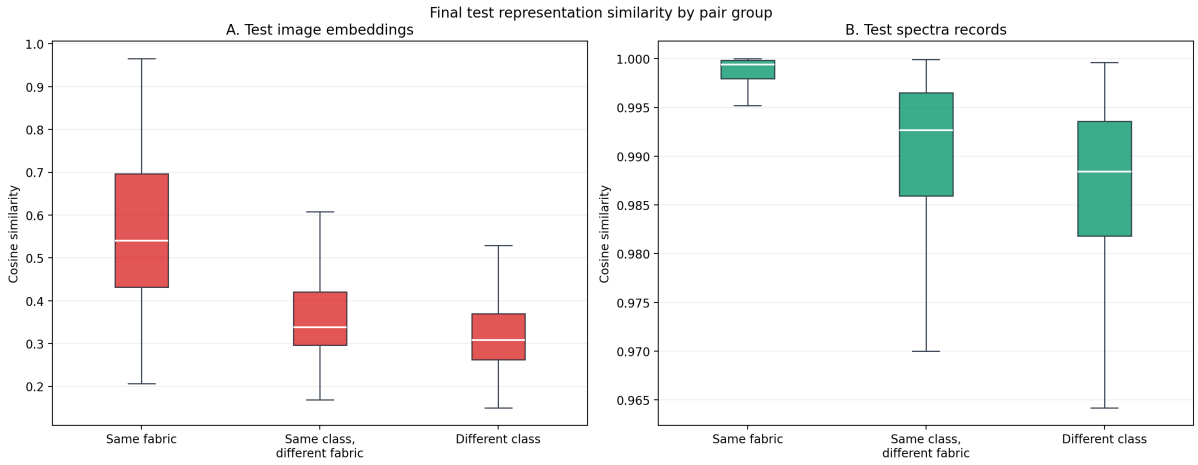


Fig. 14. Final test image and spectra representation similarity by pair group.

4.3.4 Grad-CAM Heatmaps.

Figure 15 shows Grad-CAM heatmaps for nine correctly predicted test images from the three largest CNN classes in the test split. The highlighted regions are local and usually align with visible texture structures in the microscopy images.

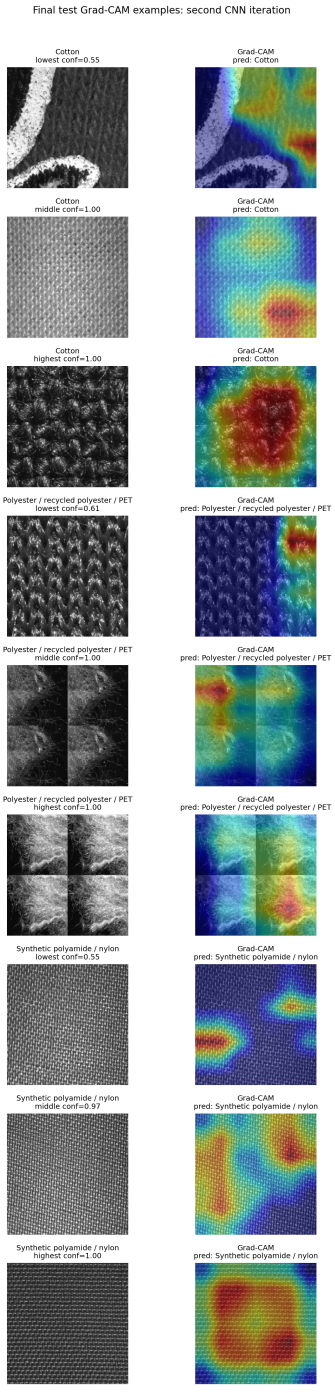


Fig. 15. Grad-CAM heatmaps for selected correctly predicted test microscopy images in the final test evaluation.

5 Discussion

5.1 Contribution and Intended Use

The background review shows that related work usually addresses one part of fabric characterisation: image-based textile classification, spectroscopy-based identification, or dataset-level comparison. This thesis instead combines microscopy image embeddings and NIR spectra as separate test representations in one decision-support workflow. The prototype supports two modes: reference comparison, where a fabric is compared with a selected stored reference, and composition estimation, where composition is estimated from similar stored fabrics with known fibre-composition labels.

The main contribution is therefore an inspectable multimodal architecture for fabric characterisation. Rather than hiding the evidence behind one final score, the prototype keeps image-only output, spectra-only output, warnings, and the combined result visible. This design choice matters because the evaluated modalities behave differently: spectra-only retrieval gives the strongest composition-estimation result, while the fine-tuned image representation provides a separate, weaker but still informative source of evidence. Keeping the outputs side by side lets a specialist see where a result came from instead of trusting a single merged number.

The prototype iterations shaped this contribution. The first low-fidelity design focused on composition estimation; the second added reference comparison after Responsible IT feedback. This separates two user needs: checking a fabric against a known reference, and estimating composition when no direct reference is available. Error flows and visible per-test outputs support this separation by keeping missing evidence, low similarity, and modality disagreement inspectable.

The architecture is designed so that a textile specialist validating a composition claim can weigh the image and spectra evidence independently rather than relying only on a combined score. Whether this design leads to more accurate decisions in practice has not yet been evaluated with users. Similarly, while the architecture could in principle be extended toward routing decisions for recyclers or sorters, routing was not tested in this thesis and would require further evaluation under operational conditions.

5.2 Individual and Societal Impact

These design choices carry implications beyond technical performance. The most direct users are textile specialists, recyclers or sorters, and organisations preparing product information. For these actors, the prototype structures interpretation by showing image-only, spectra-only, and combined outputs side by side, and warns when a test is missing, similarity is low, or modalities disagree. This can guide decisions about retesting, adding another test, manual review, validating a DPP composition claim, or routing fabrics for recycling. For a textile specialist, this changes the task from interpreting separate test results in isolation to reviewing a structured comparison in which each modality, warning, and suggested next step is visible in one place.

Inspectability, however, does not eliminate the risk of overtrust. If users treat the combined output as more certain than the evidence supports, an incorrect estimate could influence validation, product-information, or recycling decisions. The prototype therefore requires a human-in-the-loop: it provides evidence and warnings, but responsibility for validation remains with the user.

Consumers are affected more indirectly. They are unlikely to inspect embeddings or spectra similarities themselves, but they depend on the quality of information entering the Digital Product Passport. For regulators, better characterisation support can improve traceability between a composition claim and the evidence behind it. Incorrect or overconfident estimates, however, could also support misleading product or sustainability claims.

Responsible IT, the technical stakeholder, reviewed the prototype direction and influenced the separation between reference comparison and composition estimation, weighting, and error flows. Broader review with textile specialists and other stakeholders is still needed.

5.3 Model Quality and Limitations

Using the CRISP-ML(Q) view that model quality includes more than predictive performance, the evaluation considered performance, robustness, explainability, scalability, and model complexity [55].

Fine-tuning the final DenseNet201 feature block improved validation performance compared with the frozen iteration. The test result is lower than validation but remains above the majority-class baseline. The validation result shows that fine-tuning improves the selected image representation, while the held-out test split gives a more cautious estimate of generalisation.

Class-level conclusions are limited because the dataset is small and imbalanced. Cotton and polyester / recycled polyester / PET have the clearest evidence; minority-class results are sensitive to individual fabrics. NIR-SORT is useful because it links microscopy images, spectra, and fibre-composition metadata for the same fabric samples, but the validation and test splits contain few fabrics per class. The CNN classification task also uses dominant-fibre labels, which simplifies classification and leaves full blend similarity to the retrieval and composition-estimation evaluation. Fabrics with no dominant fibre or tied dominant fibres are excluded from CNN classification metrics, though they remain part of the broader fabric-characterisation problem.

The image model inherits a domain gap from ImageNet pretraining: DenseNet201 is trained on natural images, while the target data consists of textile microscopy images. Fine-tuning adapts the representation, but the small dataset limits how far adaptation can go without overfitting risk.

For composition estimation, spectra-only retrieval is strongest on both validation and test. Image-only retrieval outperforms the training-mean baseline, confirming that image embeddings carry composition-related information, but it remains weaker than spectra-only retrieval. The combined image + spectra configurations sit between the two single-modality results. This supports the interface choice to show separate modality outputs next to the combined estimate: the combined result can obscure which modality is driving the prediction. The findings also support the requirements for per-test outputs, combined outputs, uncertainty warnings, and human-in-the-loop decision support.

Representation-similarity checks show useful but imperfect structure. For image embeddings, same-fabric pairs have the highest median similarity, followed by same-class different-fabric pairs and different-class pairs, though the groups overlap. Spectra show the same ordering, but cosine similarities are concentrated near 1.0. This difference limits reference comparison: image and spectra similarities operate in different score ranges, so per-test normalisation or calibration is needed before a combined reference-comparison score can be interpreted reliably.

Robustness was assessed through representation-similarity checks, and the image-augmentation check documented in the project wiki [7]. The prototype was not tested on other textile datasets, instruments, or real operational conditions. Broader robustness therefore remains future work, and the findings cannot yet be assumed to hold across different datasets or instruments.

The Grad-CAM check showed that fine-tuning changed the highlighted image regions: the frozen model produced more diffuse heatmaps, while the fine-tuned model produced more localised regions that often followed visible texture structures. This supports the interpretation that fine-tuning changed how the CNN uses the images, although the heatmaps do not prove chemical meaning.

The scalability check shows approximately linear CNN inference time on the NVIDIA GeForce GTX 1060 6GB. It shows CNN inference after preprocessing, while full workflow performance also depends on preprocessing and vector retrieval. The result should therefore be read as evidence about the CNN inference stage only, not about full-system scalability.

Several prototype choices remain untuned. The neighbour count $k = 3$ may not be optimal for image and spectra retrieval. The highest CNN softmax output is an uncalibrated image-neighbour confidence proxy, and spectra confidence is unavailable.

5.4 Future Work

Future work falls into four areas: data, retrieval and fusion, user evaluation, and system-level evaluation.

5.4.1 Data.

A larger and more balanced dataset, with more minority-class fabrics and within-class variation, would strengthen evaluation of model choices, fine-tuning choices, and per-class performance. This is the most important limitation, because many of the other open questions, such as optimal k, confidence calibration, and per-class reliability, depend on having enough data per class to draw stable conclusions.

5.4.2 Retrieval and fusion.

The retrieval and fusion settings also need further development. The best neighbour count k may differ between image and spectra retrieval, and alternative fusion strategies should be compared. For reference comparison, image and spectra similarities need per-test normalisation before they can be combined into an interpretable score. Confidence values, especially for the CNN where the highest softmax output is used, also need calibration.

5.4.3 User evaluation.

A walkthrough with a software-development participant provided feedback on prototype understandability rather than end-user validation. The participant understood the two modes and found the flow clear, but noted that the BWM weighting process requires domain knowledge to judge the relative importance of the tests, and that warnings, visual states, and action labels could be clearer [14]. Future evaluation should involve textile specialists and test whether users understand weights, confidence, per-test outputs, warnings, next steps, and the distinction between reference comparison and composition estimation.

5.4.4 System evaluation.

Finally, the prototype should be evaluated as a complete system, since end-to-end performance depends on preprocessing, vector retrieval, interface behaviour, and use within a DPP workflow.

6 Conclusion

This thesis investigated how an AI-supported decision system using multimodal tests can contribute to fibre characterisation while handling uncertainty and supporting textile specialists in a Digital Product Passport context. The prototype structures microscopy-image and NIR-spectra evidence as separate test outputs before fusion. It supports reference comparison, composition estimation, warnings for missing or conflicting evidence, and review of the underlying per-test outputs.

The results provide a cautious answer to the research question. Fine-tuning the final DenseNet201 feature block improved validation weighted-F1 to 0.906, while locked-test weighted-F1 was 0.750, still above the majority-class baseline. This suggests that image evidence is useful, but not stable enough to treat as conclusive. For composition estimation, spectra-only retrieval performed strongest, image-only retrieval improved over the training-mean baseline, and combined estimates fell between the separate modalities. This supports showing per-test outputs next to the combined result, because fusion can hide differences between image and spectra evidence.

The main contribution is an inspectable multimodal decision-support architecture for DPP-oriented fabric characterisation. It combines late fusion, similarity-based retrieval, and BWM weighting, and is evaluated beyond performance through composition error, representation similarity, Grad-CAM explainability, CNN inference scalability, and model complexity.

The main limitations are the small and imbalanced dataset, dominant-fibre simplification, ImageNet to microscopy domain gap, uncalibrated softmax confidence, unnormalised reference-comparison scores, and untuned retrieval settings. Future work should use larger datasets, calibrate confidence and reference scores, tune k,

thresholds, and evaluate whether textile specialists can use the outputs, warnings, and suggested next steps appropriately in DPP workflows.

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A Composition Estimation Happy Flow

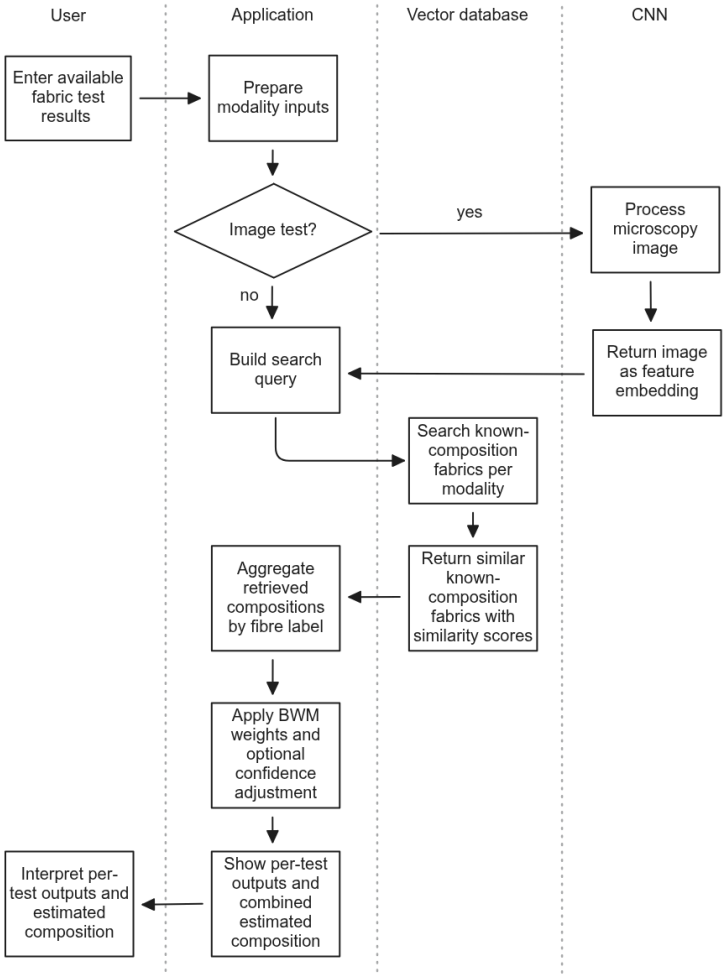


Fig. 16. Happy flow for the composition-estimation route.

B Reference Comparison Happy Flow

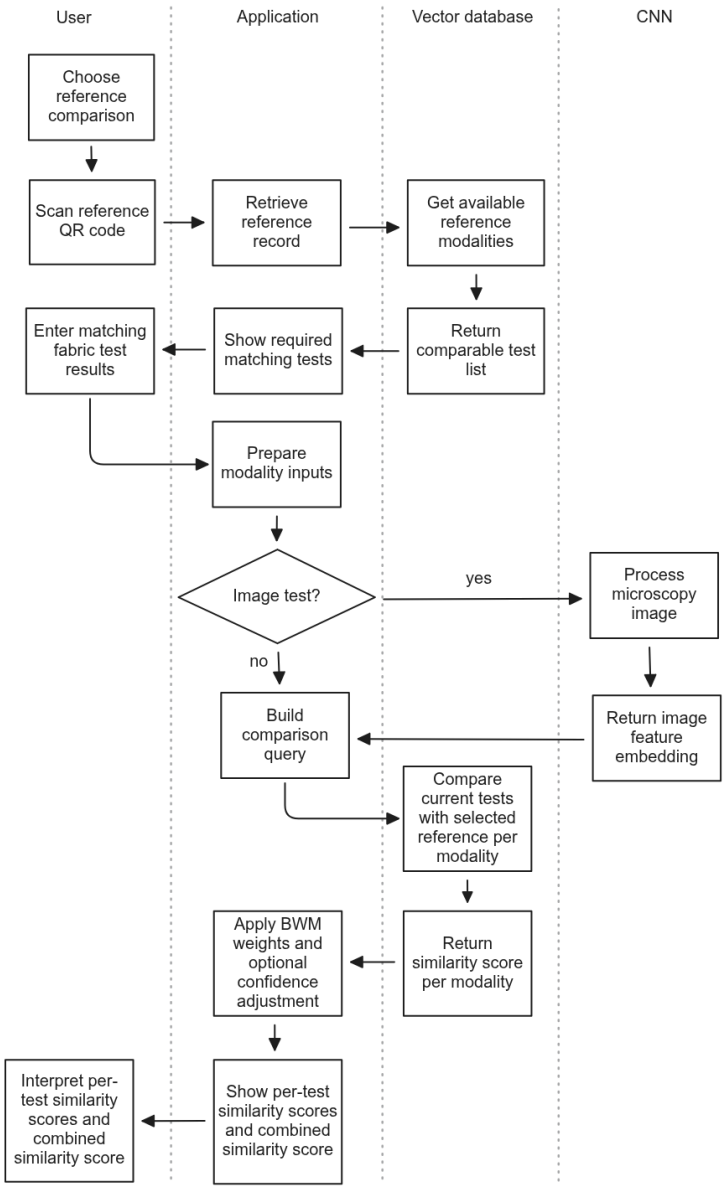


Fig. 17. Happy flow for the reference-comparison route.

C Error Flow Diagrams

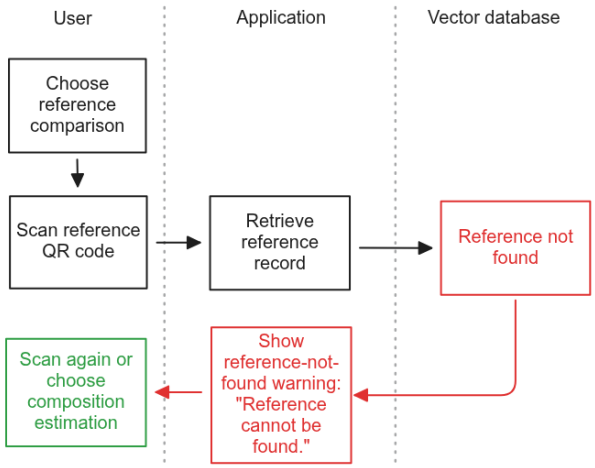


Fig. 18. Error flow for a reference fabric that cannot be found.

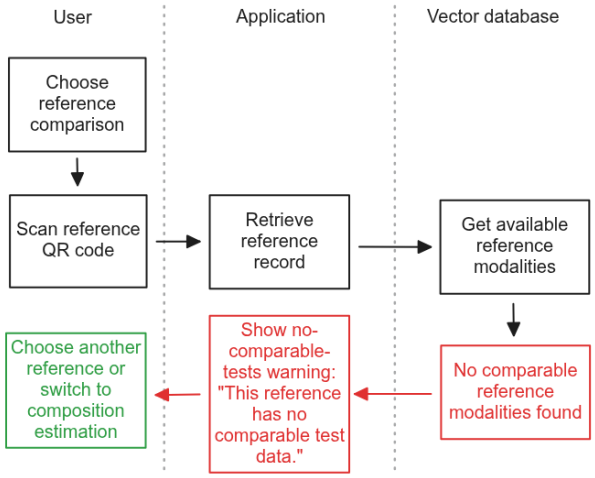


Fig. 19. Error flow for a reference fabric without matching test data.

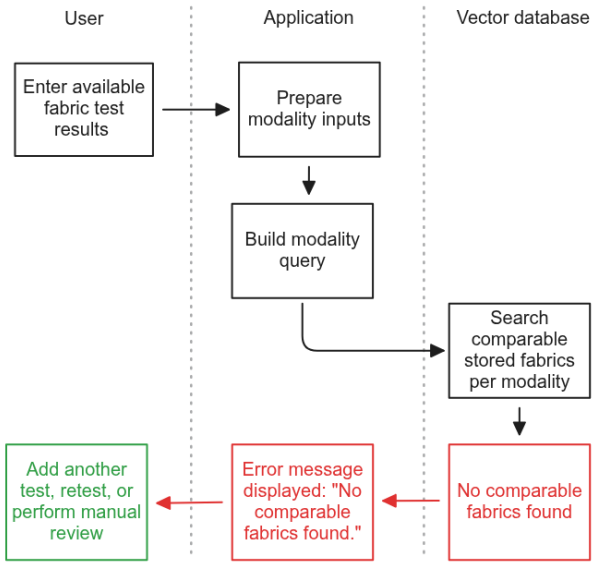


Fig. 20. Error flow for unavailable comparable stored fabrics.

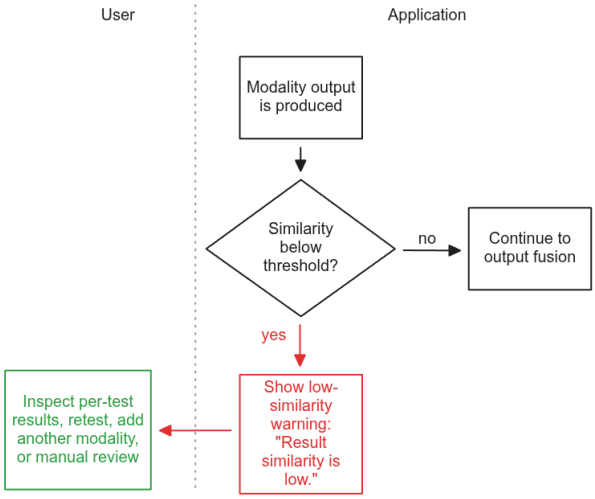


Fig. 21. Error flow for low-similarity test output.

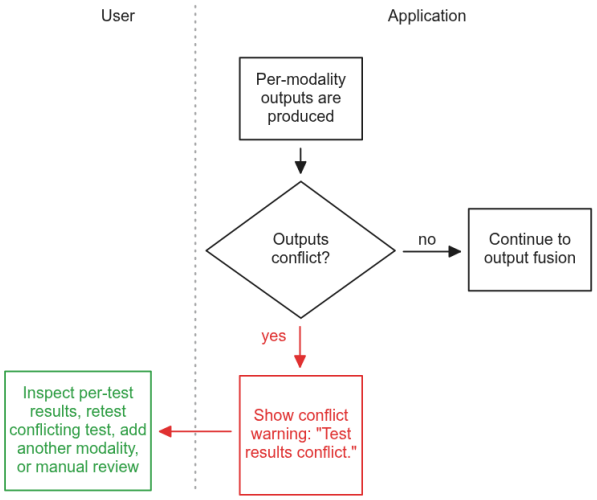


Fig. 22. Error flow for conflicting modality outputs.